Daniela Andrea HURTADO LANGE

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Educational background

Ph.D. in Operations Research (Dec. 2021)
Georgia Institute of Technology – Atlanta, Georgia; Advisor: Prof. Siva Theja Maguluri

Master of Science in Industrial Engineering (Jul. 2016)
Pontificia Universidad Católica de Chile – Santiago, Chile; Advisor: Prof. Pedro Gazmuri

Industrial Engineer with Diploma in Math Engineering (Jul. 2016)
Pontificia Universidad Católica de Chile – Santiago, Chile; *Maximum distinction*

Bachelor in Engineering (Dec. 2013)
Pontificia Universidad Católica de Chile – Santiago, Chile; *With distinction*

Work experience

Kellogg School of Management, Northwestern University
Assistant Professor of Operations (Sep. 2024 to date)
Drake Scholar, Donald P. Jacobs Scholar (Jul. 2023 – Aug. 2024)

Mathematics Department, William & Mary
Assistant Professor of Mathematics (Jan. 2022 – Jun. 2023)

Amazon.com Services LLC
Research Scientist Intern (May – Aug. 2021)

Industrial & Systems Engineering, Georgia Institute of Technology
Graduate Research Assistant and Graduate Teaching Assistant (Aug. 2016 – Dec. 2021)

Industrial & Systems Engineering, Pontificia Universidad Católica de Chile
Lecturer (Mar. – Jul. 2016)

Alcon Chile
Summer Intern (Dec. 2012 – Feb. 2013)

Research interests

Applied Probability; Queueing theory; Stochastic Processing Networks; Heavy-traffic analysis

Publications

Working papers

[14] **Hurtado-Lange, D.** and Grosof, I. Heavy-traffic and state space collapse analysis for parallel-server networks with Markov-modulated inputs.

[13] van Kempen, S., Varma, S. M., **Hurtado-Lange, D.**, and Sloothaak, F. A priority policy for a skill-based parallel-server network: δ fluid-optimality and heavy-traffic guarantees.

Under revision

[11] **Hurtado-Lange, D.** and Grosof, I. (2026). Markov modulated JSQ in heavy traffic via the Poisson equation.

[10] Grosof, I. and **Hurtado-Lange, D.** (2025). Outperforming multiserver SRPT at all loads. Journal version of conference paper [9].

[8] Jhunjhunwala, P., **Hurtado-Lange, D.**, and Maguluri, S. T. (2023). Exponential tail bounds on queues: A confluence of non-asymptotic heavy-traffic and large deviations. Journal version of conference paper [7].

Published and forthcoming

[12] Varma, S. M., Jhunjhunwala, P., **Hurtado-Lange, D.**, and Maguluri, S. T. (2026). Transform method for stochastic processing and matching networks. In: *TutORials in Operations Research*. INFORMS. 2026 INFORMS TutORials in Operations Research. To appear.

[9] Grosof, I. and **Hurtado-Lange, D.** (2026). Outperforming multiserver SRPT at all loads. In: *Abstracts of the 2026 ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems*. SIGMETRICS Abstracts '26. USA: Association for Computing Machinery, pp. 64–66.

[7] Jhunjhunwala, P., **Hurtado-Lange, D.**, and Maguluri, S. T. (2023). Exponential tail bounds on queues. In: *ACM SIGMETRICS Performance Evaluation Review*. Vol. 51. 2, pp. 24–26.

[6] **Hurtado-Lange, D.**, Varma, S. M., and Maguluri, S. T. (2022). Logarithmic heavy traffic error bounds in generalized switch and load balancing systems. In: *Journal of Applied Probability* 59.3, pp. 652–669.

[5] **Hurtado-Lange, D.** and Maguluri, S. T. (2022). A load balancing system in the many-server heavy-traffic asymptotics. In: *Queueing Systems* 101.3, pp. 353–391.

[4] **Hurtado-Lange, D.** and Maguluri, S. T. (2022). Heavy-traffic analysis of queueing systems with no complete resource pooling. In: *Mathematics of Operations Research* 47.4, pp. 3129–3155. ⋆ Second prize, 2020 Junior Faculty Interest Group (JFIG) Paper Competition. Journal version of conference paper [2].

[3] **Hurtado-Lange, D.** and Maguluri, S. T. (2021). Throughput and delay optimality of power-of-d choices in inhomogeneous load balancing systems. In: *Operations Research Letters* 49.4, pp. 616–622.

[2] **Hurtado-Lange, D.** and Maguluri, S. T. (2020). Heavy-traffic analysis of the generalized switch under multidimensional state space collapse. In: *ACM SIGMETRICS Performance Evaluation Review*. Vol. 47. 2, pp. 36–38. Acceptance rate 15%. A preliminary version appeared at the MAMA workshop, ACM SIGMETRICS 2019 (abstract).

[1] **Hurtado-Lange, D.** and Maguluri, S. T. (2020). Transform methods for heavy-traffic analysis. In: *Stochastic Systems* 10.4, pp. 275–309.

Selected presentations

A selection of recent talks appears below. Slides and videos (when recorded), together with a complete list of seminars and posters, are available on my website.

Workshops and conference presentations

- **Transform Methods for Markov-Modulated Queues**
  November 2025 – YEQT 2025 – Eindhoven, Netherlands
  October 2025 – INFORMS 2025 – Atlanta, GA
  July 2025 – APS Conference 2025 – Atlanta, GA
  October 2024 – INFORMS 2024 – Seattle, Washington (introduction to the problem and methodology)
- **Exponential Tail Bounds on Queues**
  September 2024 – Allerton conference at UIUC – Urbana, Illinois
- **Efficient Algorithm For Transient Analysis Of Cox/M/m Queues**
  August 2024 – CanQueue 2024 – Edmonton, Alberta, Canada
  October 2023 – INFORMS 2023 – Phoenix, Arizona
- **Queue Length Behavior in Load Balancing Systems Under Power-of-d Choices: Many-Server Heavy-Traffic Regime**
  October 2021 – INFORMS 2021 – Anaheim, California and online
- **Load Balancing System under JSQ and Power-of-d: Many-Server Heavy-Traffic Asymptotics**
  November 2020 – INFORMS 2020 – Held online
- **Heavy-Traffic Analysis of the Generalized Switch under Multidimensional State Space Collapse**
  June 2020 – SIGMETRICS 2020 – Held online
  October 2019 – INFORMS 2019 – Seattle, Washington
  June 2019 – MAMA Workshop at ACM SIGMETRICS 2019 – Phoenix, Arizona (preliminary version)
- **Transform Methods for Heavy-Traffic Analysis**
  June 2019 – INFORMS-APS 2019 – Brisbane, Australia

Seminar presentations

- **Transform Method for Markov-Modulated Queues**

December 2025 – SNAPP seminar – Held online

- **Transform Method for Heavy-Traffic Analysis: A Tutorial**
  December 2024 – Math Department, William & Mary – Williamsburg, Virginia
- **Performance Analysis of Data Center Networks: Drift Method and Transform Techniques**
  October 2024 – ORIE, Cornell University – Ithaca, New York
  July 2024 – Industrial and Systems Engineering Department, Pontificia Universidad Católica de Chile – Santiago, Chile
  October 2022 – Kellogg School of Business, Northwestern University – Evanston, Illinois
- **Minimizing Delay in Supermarket-Checkout Systems**
  April 2022 – Math department at William & Mary

Selected scholarships and awards

- Sigma Xi Best Ph.D. Thesis Award (Fall 2022) – Georgia Tech
- Ed Iacobucci Fellowship for Excellence in Research: Applied Probability & Simulation (Spring 2021) – Georgia Tech, ISyE
- Honorable mention, Alice and John Jarvis Research Award (Spring 2020) – Georgia Tech, ISyE
- BECAS CHILE graduate scholarship (2018–2021) – Chilean government

Teaching experience

Instructor of record

Kellogg School of Management, Northwestern University – MBA

- **Core Operations Management (OPNS 430)**
  Full-time MBA (Spring 2024), MiM (Winter 2025–26), Evening & Weekend (Winter 2025–26)

William & Mary, Mathematics

- Operations Research: Deterministic Models (MATH 323); Undergraduate level; Fall 2022, Spring 2022
- Elementary Probability & Statistics (MATH 106); Undergraduate level; Fall 2022
- Analysis of Stochastic Networks (CSCI 688); Master's level; Spring 2022
- Pre-calculus boot camp (co-taught mini-course); Summer 2022

Georgia Tech / PUC Chile (as a graduate student)

- Engineering Optimization (ISYE 3133), Hybrid class; Undergraduate level; Fall 2020
- Pre-academic physics, PUC (online mini-course); August 2020
- Stochastic Models (ICS 2123), Undergraduate level; PUC Chile; Mar.–Jul. 2016

Teaching assistant

- Georgia Tech, ISyE (*2016–2021*):
  Stochastic Processes I (PhD), Statistics & Applications (MS), Special Topics: Math of OR (PhD), Probabilistic Models (MS), Simulation Analysis & Design (UG).
- PUC Chile (*2010–2015*):
  Graduate and undergraduate courses including Advanced Stochastic Models, Marketing, Stochastic Models, Probability & Statistics, Linear Algebra, and Introduction to Calculus.

Mentoring

- Research mentoring of Master's, undergraduate, and junior PhD students at William & Mary and Georgia Tech, including the SURE program (Summers 2018–2019).
- Academic mentoring at PUC Chile through Talent + Inclusion (2014–2015) and CARA UC (2013–2015), the latter as coordinator of math mentors.

Professional service

Professional and organizational roles

- Applied Probability Society (APS) Council – *Jan. 2026 to date*
  * Nominated by 2025 council members, voted by INFORMS APS members
- Webmaster for INFORMS Applied Probability Society (APS) – *Jan.-Dec. 2025*

- Website and communications chair for 2025 INFORMS APS Conference – *Jun. 2024 - Jul. 2025*
- SIGMETRICS Webmaster – *Dec. 2019 to Dec. 2023*
- ISyE Student Seminars Organization Committee – *2018–2021*
  Industrial and Systems Engineering Department at Georgia Institute of Technology – Atlanta, Georgia

Conference and Workshop Sessions

- Chair in session named "Top-notch Applications of Queueing Theory"
  INFORMS 2025 – *Oct. 2025* – Atlanta, GA, USA
- Organizer and chair of "Women in APS" panel
  APS 2025 – *Jul. 2025* – Atlanta, GA, USA
- Chair in session named "New Developments in Queueing and Learning"
  APS 2025 – *Jul. 2025* – Atlanta, GA, USA
- Chair in session named "Queueing and Applications"
  INFORMS 2024 – *Oct. 2024* – Seattle, WA, USA
- Chair in session named "Queueing Theory and Applications"
  INFORMS 2023 – *Oct. 2023* – Phoenix, AZ, USA
- Chair in session named "Recent Advances in Parallel-Server Systems" in the Applied Probability Society (APS) cluster
  INFORMS 2022 – *Oct. 2022* – Indianapolis, IN, USA
- Chair in session named "Recent Advances in Parallel-Server Systems"
  2022 CORS/INFORMS International Conference – *Jun. 2022* – Vancouver, Canada
- Chair in session named "Advances on Queueing and Learning" in the Applied Probability Society (APS) cluster
  INFORMS 2021 – *Oct. 2021* – Anaheim, CA, USA
- Co-chair in session named "Asymptotic Analysis of Stochastic Processing Networks" in the APS cluster
  INFORMS 2020 – *Nov. 2020* – Held online

Reviewer

Applied Probability Journals, Annals of Operations Research (ANOR), Institute of Industrial and Systems Engineers (IISE) Transactions, Management Science, Operations Research Letters, Operations Research, Queueing Systems (QUESTA), Performance Evaluation (PEVA), Stochastic Models (LSTM), Stochastic Systems, IEEE/ACM Transactions on Networking.

Professional associations

- INFORMS: Applied Probability Society (APS), Junior Faculty Interest Group (JFIG), Minority Issues Forum (MIF), WORMS Forum
- Institute of Mathematical Statistics (IMS): Stochastic Networks
- Sigma Xi Research Society (2022-2023).
  * Membership granted as part of the Sigma Xi Best Ph.D. Thesis Award in 2022

Languages

- Spanish: Native speaker
- English: Full professional proficiency